

Baigang Chen

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Education

University of Minnesota, Twin Cities

M.S. Mathematics, co-advised by Prof. Nick Hopper and Prof. Andrew Odlyzko

Starting from Fall 2025

University of Washington, Seattle

B.S. in Mathematics

Fall 2022 - Spring 2025

Honors and Awards

Annual Dean's List of the University of Washington, Seattle.

2023 – 2024

Publications and Pre-prints

- [1] **Baigang Chen** and Dongfang Zhao. *Reliable Non-Leveled Homomorphic Encryption for Web Services*. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, pp. 3171–3182, ACM, 2026. doi:10.1145/3774904.3792476.
- [2] **Baigang Chen** and Nicholas Hopper. *G-Lox: Group-Adaptive, Privacy-Preserving Bridge Distribution with Two-Party Computation*. <https://arxiv.org/abs/2606.19620>

Research Experiences

ZKP in LLM

Advisor: James Petrie

July 2026 – Present

Organization: MARS V, Oxford, UK

Description: Based on Verlnf, Ligerio, and zkLLM. This individual project mainly focuses on protocol design so that the prover doesn't need to reveal the model architecture.

Bidirectional Fully Encrypted Protocols

Advisor: Nick Hopper, University of Minnesota, Twin Cities

July 2026 – Present

Description: Motivated by the observation that a trivial composition of two unidirectional fully encrypted protocols may still expose protocol-identifying traffic patterns, we extend the unidirectional setting to a bidirectional fully encrypted protocol with public emission and close schedules. In each direction, the protocol transmits dummy messages whenever no application data are available, thereby preserving the prescribed traffic shape.

Outcome: First author paper under peer review at NDSS 2027.

Order Revealing Homomorphic Encryption

Advisor: Dongfang Zhao, University of Washington, Seattle

Oct 2025 – April 2026

Co-advisor: Nick Hopper, University of Minnesota, Twin Cities

Description: Investigated order-revealing/order-preserving comparison for encrypted data, focusing on TFHE-style programmable bootstrapping to compute sign/compare over ciphertexts. Using a vFHE-style construction for proof-gate-based compare pipelines.

Outcomes: Paper under review, artifact available at https://github.com/Baigang0000/Order_reveal_FHE

Revisiting rBridge

Advisor: Nick Hopper, University of Minnesota, Twin Cities

Sep 2025 – April 2026

Description: Designed and analyzed **G-Lox**, a privacy-preserving bridge distribution system extending Lox with group-adaptive bridge assignment and a two-server architecture that hides group state and assignment logic behind MPC-backed state services. Evaluated its privacy and system tradeoffs against Sybil-driven enumeration and censor-driven bridge discovery.

Outcomes: Paper under review at NDSS 2027, artifact available at <https://github.com/Baigang0000/G-lox>

Toward Non-leveled Approximating FHE

Advisor: Dongfang Zhao, University of Washington, Seattle

May 2025 – Nov 2025

Lab: HPDIC

Description: We constructed a non-leveled prototype of the CKKS homomorphic encryption scheme. This design fixes a single module and applies ECC to it. Based on this work, we are also working on error-correction layers for fully homomorphic encryption and general polynomially embedded data.

Outcomes: We present our poster at USENIX 2025 and publish the paper in WWW 2026.

Managing Delays in Public Transit Systems

Advisor: Giovanni Inchiostro, University of Washington, Seattle

Mar 2025 – Jun 2025

Description: In this project, our group analyzed problems and algorithms used for managing delays in public transit systems. Drawing from the book *Optimization in Public Transportation* by Anita Schöbel, we developed models and tested algorithms on small examples to implement our algorithms on real-world data from Seattle's public transportation

network.

Outcomes: We presented our poster at the University of Washington, WXML project presentation room. The coded project is available at <https://github.com/Inc-G/Public-transportation>.

Fourier Analysis and Applications

Advisor: F. Alberto Grünbaum, University of California, Berkeley

May 2024 – Jul 2024

Description: Professor Grünbaum's exclusive summer program: an intensive review of Fourier analysis and mathematical physics, including the Heisenberg uncertainty principle and heat equation. Working on applying FFT style tricks to the AKS prime test.

Outcomes: Individual author research report: *Improving AKS Prime Test by Fourier Analysis Approach*. Publication declined, 2025.

Seminars

Reading Seminar on Étale Cohomology

Supervisor: Kai-Wen Lan

Sep 2025–Dec 2025

Present: Overview on Weil Conjectures (Appendix C to Hartshorne's Algebraic Geometry).

Anonymous Communications Theory

Supervisor: Megumi Ando, Tufts University

Jan 2025 – May 2025

Description: Graduate seminar presenting cryptography-style definitions of anonymity and studying state-of-the-art techniques for achieving provable anonymity.

Burnside Problem

Supervisor: Be'eri Greenfeld, University of Washington, Seattle

Mar 2025 – June 2025

Special topic in advanced undergraduate mathematics (UW Math 480 SP 25). A seminar-style lecture on the Burnside Problem, which was solved in stages by Golod and Shafarevich, Adian-Novikov and Olshanskii, and Zelmanov, linking p-groups, Lie algebras, and polynomial identities, impacting group theory and algebraic number theory.

Mathematics of Public Key Cryptography

Supervisor: Michael Naehrig and Natalie Naehrig, University of Washington, Seattle

Mar 2024 – June 2024

Special topic in advanced undergraduate mathematics (UW Math 480 SP 24). A seminar-style lecture introduces students to the mathematical concepts used in public key cryptography and their real-world applications.

Reading Projects

Direct Reading Projects

Oct 2025 – Jan 2026

Role: Mentor

Materials: A Graduate Course in Applied Cryptography by Dan Boneh and Victor Shoup.

Outcome: Mentee presents baby-step-giant-step algorithm at the DRP presenting room.

Conference Attended/Scheduled

The ACM Web Conference (WWW 2026)

July 2026

Paper accepted: *Reliable Non-Leveled Homomorphic Encryption for Web Services*.

USENIX Security Symposium (USENIX Security 2025)

Aug 2025

Poster Presentation: *Binary CKKS: A Non-Leveled Approximate-Number FHE Scheme*. Baigang Chen and Dongfang Zhao, University of Washington.

Fields of Interest

Cryptography. Applied Cryptography. Censorship resistance.